

PRASANNA VARDHMAN GEBISE

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Education

Vishwakarma Institute of Technology, Pune

May 2025

Bachelor of Technology in Electronics and Telecommunication (GPA: 8.07 / 10.00)

Pune, Maharashtra

- **Relevant Coursework:** Data Structures and Algorithms, OOPS, OS, Digital Design
- **Languages:** Python, C++

Technical Skills

Robotics/Autonomy: ROS, ROS2, Gazebo, SLAM (monocular/stereo Visual & Visual-Inertial SLAM), Sensor Fusion (Kalman Filter/EKF), State Estimation, GNSS-denied Navigation, Finite State Machines, LQR & Optimal Control

UAV/Drone: UAV Design & Fabrication, VTOL Aircraft Design, Drone Firmware, Avionics, Aerodynamics, Flight Testing

Manipulators & Mechatronics: Cable-Driven Parallel Manipulators, Forward Kinematics, Cable Tension Optimization (Quadratic Programming), Holonomic Robot Control

CAD/Structural: CAD Modelling, Structural Analysis, CFD Analysis

CV/SLAM Stacks: OpenCV, ORB-SLAM3, VINS-Fusion, DM-VIO, OKVIS2-X, OpenSfM, COLMAP

Simulation & Flight: Gazebo, Ardupilot, Mission Planner, Pixhawk, xflr5

Embedded/Compute: Arduino, Raspberry Pi 5, Jetson Orin NX, sbRIO

Engineering/Analysis: LabVIEW, MATLAB, Octave, Fusion360, Ansys

Dev Tools: VSCode, GitHub

Experience

VTOL Aviation India, Pvt. Ltd.

Jan 2026 – Current

Computer Vision Engineer

Navi Mumbai, Maharashtra

Working on GNSS denied localization for UAVs. Algorithms tested on indoor data with custom camera-imu setup- ORBSLAM3, VINS-FUSION, DM-VIO and OKVIS2-X. Successfully tested and validated ORBSLAM3 in mono-only mode on outdoor data with 15 m mean error from the ground truth. Cross-View Geo-Localization for global drift correction of the visual/visual-inertial odometry.

Zebu Intelligent Systems

Jun 2025 – Nov 2025

Robotics Software Intern

Hyderabad, Telangana

Worked on firmware for drones, including implementation and testing of the following in simulation (Gazebo), as well as hardware deployment using onboard computer on drone. Had experience using both Raspberry pi 5 and Jetson Orin NX for hardware deployment.

- Kalman Filter for fusing multiple sensors like GPS, Camera and Rangefinder
- Finite State machine for smooth transition of drone while performing multiple tasks
- Visual SLAM and odometry using ORB-SLAM with monocular and stereo camera and IMU configurations
- Scene Reconstruction using structure from motion approaches like open-sfm and colmap
- OpenCV trackers and algorithms for feature detection and description like ORB, SIFT, SURF

Human Centered Robotics Lab, IIT Gandhinagar

Aug 2024 – Apr 2025

Research Intern

Gandhinagar, Gujarat

- Developed a cable driven parallel manipulator testbed for upper limb rehabilitation. My practical contributions include sensor interface, driving motors and development of a controller for maintaining desired forces on end effector which was deployed and tested in real-time on sbRIO using LabVIEW. Theoretical contributions include implementation of forward kinematics solution for over-constrained cable driven parallel manipulator using analytical approach and optimization of cable tensions using quadratic programming in LabVIEW.

e-Yantra, IIT Bombay
Summer Intern

May 2024 – Jun 2024
Mumbai, Maharashtra

- Developed an onboard LQR (Linear quadratic regulator) controller for a quadcopter using the feedback from a VIO (Visual Inertial Odometry) pipeline for state estimation of the drone indoors or at places where GPS is inaccessible.

e-Yantra, IIT Bombay
Summer Intern

May 2023 – Jun 2023
Mumbai, Maharashtra

- Developed an efficient Vertical Takeoff and Landing (VTOL) aircraft from scratch. Aerodynamic stability analysis in xflr5, 3D modelled the VTOL in Fusion360, Static structural and CFD analysis in Ansys. Fabricated the VTOL, integrated on board electronics and flight testing.

Projects

All my projects are showcased here-
Portfolio

- GNSS denied UAV navigation framework
- Visual and Visual-Inertial SLAM
- Drone-in-a-box
- Cable-Driven Parallel Manipulator Testbed
- Onboard LQR with VIO for Quadcopters
- ADHIC – Autonomous Disinfectant Holonomic Interactive Cobot
- Lunar Scout – Self-Balancing Bike
- HolAbot – Holonomic Art Robot
- Hybrid VTOL – Design and Fabrication

Achievements

eYSIP 2023 and 2024 Summer Intern at IIT Bombay.
AIR 3 in eYantra robotics competition 2022-23 held at IIT Bombay.
AIR 4 in eYantra robotics competition 2023-24 held at IIT Bombay.
One patent published- ADHIC(Autonomous Disinfectant Holonomic Interactive Cobot).